

APPM 4600 — HOMEWORK # 3

1. **Definition 1** Suppose $\{p_n\}_{n=0}^{\infty}$ is a sequence that converges to p with $p_n \neq p$ for all n . If there exists positive constants λ and α such that

$$\lim_{n \rightarrow \infty} \frac{|p_{n+1} - p|}{|p_n - p|^\alpha} = \lambda$$

then $\{p_n\}_{n=1}^{\infty}$ converges to p with an order α and asymptotic error constant λ . If $\alpha = 1$ and $\lambda < 1$ then the sequence converges linearly. If $\alpha = 2$, the sequence is quadratically convergent.

Which of the following iterations will converge to the indicated fixed point x_* (provided x_0 is sufficiently close to x_*)? If it does converge, give the order of convergence; for linear convergence, give the rate of linear convergence.

(a) $x_{n+1} = -16 + 6x_n + \frac{12}{x_n}$, $x_* = 2$

(b) $x_{n+1} = \frac{2}{3}x_n + \frac{1}{x_n^2}$, $x_* = 3^{1/3}$

(c) $x_{n+1} = \frac{12}{1+x_n}$, $x_* = 3$

2. In laying water mains, utilities must be concerned with the possibility of freezing. Although soil and weather conditions are complicated, reasonable approximations can be made on the basis of the assumption that soil is uniform in all directions. In that case the temperature in degrees Celsius $T(x, t)$ at a distance x (in meters) below the surface, t seconds after the beginning of a cold snap, approximately satisfies

$$\frac{T(x, t) - T_s}{T_i - T_s} = \operatorname{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right),$$

where T_s is the constant temperature during a cold period, T_i is the initial soil temperature before the cold snap, α is the thermal conductivity (in meters² per second), and

$$\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t \exp(-s^2) ds$$

Assume that $T_i = 20$ [degrees C], $T_s = -15$ [degrees C], $\alpha = 0.138 \cdot 10^{-6}$ [meters² per second]. It is convenient to use `scipy` to evaluate the `erf` function.

For parts (b) and (c), run your experiments with a tolerance of $\epsilon = 10^{-13}$.

- (a) We want to determine how deep a water main should be buried so that it will only freeze after 60 days exposure at this constant surface temperature.
Formulate the problem as a root finding problem $f(x) = 0$. What is f and what is f' ?
Plot the function f on $[0, \bar{x}]$, where $\bar{x}$ is chosen so that $f(\bar{x}) > 0$.
- (b) Compute an approximate depth using the Bisection Method with starting values $a_0 = 0$ [meters] and $b_0 = \bar{x}$ [meters].
- (c) Compute an approximate depth using Newton's Method with starting value $x_0 = 0.01$ [meters].
What happens if you start with $x_0 = \bar{x}$?

3. Let $f(x)$ denote a function with root α of multiplicity m .
- (a) Write down a formal mathematical definition of what it means for α to be a root of multiplicity m of $f(x)$.
 - (b) Show that Newton's method applied to $f(x)$ only converges linearly to the root α .
 - (c) Show that the fixed point iteration applied to $g(x) = x - m \frac{f(x)}{f'(x)}$ is second order convergent.
 - (d) What does part (c) provide for Newton's method in the case of roots with multiplicity greater than 1?
4. There are two ways of improving the convergence of Newton's method when a root has multiplicity greater than 1: Problem 2 c and apply Newton's method to $g(x) = \frac{f(x)}{f'(x)}$.

In this problem consider finding the root of the function $f(x) = e^{3x} - 27x^6 + 27x^4e^x - 9x^2e^{2x}$ in the interval $(3, 5)$.

Explore the order of convergence when applying (i) Newton's method, (ii) the modified Newton's method from class and (iii) the modified Newton's method in Problem 2. Which method do you prefer and why?